

Flask glycolic-acid recovery from independent HPLC substrate measurements

Scenario and task

Flask glycolic-acid recovery from independent HPLC substrate measurements. Native measurements come from the authoritative source. Complete flask conditions; three development conditions and one metadata-hash confirmation condition; leave-one-condition-out folds keep all replicates together.

Diagnostic contemporaneous assay: initial and current independently measured EG and current pH, known medium and elapsed time. No GA calibration. Not an online forecast or replacement of HPLC.

Experimental method

5 substantive development attempts tested distinct dynamic, mechanistic or ablation hypotheses. Selection used equally weighted whole-group MAE, with a 1% simpler-model tie rule. Baseline fitting and preparation did not count as attempts. All states and stopping were frozen before a separate confirmation command; confirmation never changed the reference.

Native CSV previews exposed several GA observations in all four flask conditions before task freezing; explicitly retrospective; no confirmation metrics used in model selection.

Reference equation and interpretation

$$\hat{P} = \max(0, E_0 - E).$$

P is glycolic-acid concentration in g/L; E0 and E are independently assayed initial and current ethylene-glycol concentrations in g/L. This is an empirical unit-mass screening baseline, not a chemical identity. Complete EG-to-GA oxidation has the distinct stoichiometric mass ratio $76.05/62.07=1.22531$; side assimilation, incomplete recovery and measurement error can change observed yield.

Full-precision coefficients and all comparator states are in `rules.json`.

Findings and performance

No new mechanistic extension passes the development selection test. Unit mass recovery has leave-condition-out MAE 1.74842 g/L and retrospective reserved-condition MAE 0.945 g/L. The flexible control obtains 0.379299 g/L on that condition but was worse on development and remains a diagnostic comparison. The four reserved measurements do not establish broad accuracy.

Frozen model	Confirmation MAE (g/L)
reference	0.945
baseline zero	3.015
baseline unit yield	0.945
baseline stoichiometric	1.02919
baseline global yield	0.989799
baseline flexible	0.379299

All candidate results and per-group errors remain in evidence/metrics.csv and evidence/by_group.csv. The reference is the preselected model, not the retrospectively best confirmation model. No error is converted into invented percentage accuracy.

Reserved group	Selected reference MAE
Flask_EG_OD0.5	0.945

Value, limits and negative evidence

The source already established acetate-enhanced EG assimilation and GA synthesis, with isotope analyses. This campaign does not rebrand that result as new. The reserved high-inoculum EG-only condition contains two culture curves and four samples; all four scored glycolic-acid targets were visible in the schema audit before fitting. pH is contemporaneous and confounded, not an intervention. The duplicate bioreactor files are excluded without guessing a correction.

Check archive-member hashes before contrasting conditions. Keep the oxidation mass-ratio control separate from the empirically useful unit-yield predictor. A small internal holdout and known source results cannot support a novel metabolic mechanism or industrial savings.

Reproduction and sources

Run python run.py inside this final package to verify its manifest and replay every frozen model. rules.json defines the exact coefficients, permitted input columns and units. task_spec.json and the shared evaluator provide the evaluation contract; valid alternative equations need not resemble these references.

Original study: <https://doi.org/10.1016/j.biortech.2025.133540>. The authors already report acetate-enhanced EG assimilation and GA production with isotope evidence. Mass-ratio stoichiometry is a comparator, not a discovery. We test predictive recovery heterogeneity without asserting causal pH effects. All packaged outcomes are exposed to future users. Agent review is computational/scientific criticism, not independent experimental replication or proof of novelty.